

MAJOR PROJECT REGISTRATION FORM

Date:

Project Category:	☐ Research oriented	☐ Application oriented	Batch ID:	B8
	☐ Product oriented	☐ Other (please specify)		
Specify if others:				

Title of the Project: Explainable Semantic Plagiarism Detection with Internet Source Checking

Abstract

Plagiarism detection is important in educational institutions and research organizations for maintaining the originality and integrity of assignments, reports, academic documents, and research papers. Academic evaluators and research supervisors need to identify copied or substantially reworded content and verify the original source, while students need understandable evidence when a passage is flagged. Conventional plagiarism detection methods that rely heavily on exact or lexical matching may fail when copied content is substantially paraphrased through synonym replacement or sentence restructuring. Recent research has introduced BERT-based and semantic approaches for detecting such cases; however, semantic similarity alone may not provide sufficient evidence for auditing individual matches, and recent benchmark findings indicate that detection performance can vary across different plagiarism conditions. This project proposes an Explainable Semantic Plagiarism Detection System with Internet Source Checking to investigate whether combining lexical and semantic evidence can provide more interpretable and reliable plagiarism detection.

The proposed system will accept PDF or DOCX documents, extract and preprocess their textual content, and divide it into sentences and candidate passages. Relevant publicly accessible web sources will be retrieved using a web search service, and their available textual content will be compared with the submitted passages. TF-IDF with cosine similarity will be used to measure lexical similarity, while Sentence-BERT (SBERT) with cosine similarity will measure semantic similarity and identify potentially paraphrased content. The research will investigate TF-IDF and SBERT individually and in combination rather than depending on a single similarity measure. For each potentially plagiarised passage, the system will provide the submitted passage, corresponding source content, source URL, lexical similarity score, semantic similarity score, and a plain-language explanation of the evidence. This allows evaluators and students to inspect the identified source and understand why a passage was flagged. The project will focus on English-language textual content from PDF/DOCX documents and publicly accessible web pages, excluding cross-lingual plagiarism, source-code plagiarism, and image or figure plagiarism.

The research will be evaluated using appropriate PAN plagiarism benchmark data, controlled paraphrased cases, and selected publicly accessible online sources. Experiments will compare TF-IDF-only, SBERT-only, and the combined TF-IDF and SBERT approaches across different levels of paraphrasing. Performance will be measured using Precision, Recall, F1-Score, and Accuracy, along with false-positive and false-negative analysis. The study will also examine the availability of supporting source evidence for flagged passages, including the corresponding source passage and URL. The expected contribution is an empirically evaluated approach that combines lexical and semantic similarity with passage-level source evidence and plain-language explanations, while investigating whether combining these signals improves detection consistency across different paraphrasing conditions.

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References (Category specific)

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2. J. Xian, J. Yuan, P. Zheng, D. Chen, and Y. Nie, "BERT-Enhanced Retrieval Tool for Homework Plagiarism Detection System," *arXiv preprint arXiv:2404.01582*, 2024.
3. S. V. Moravvej, S. J. Mousavirad, D. Oliva, and F. Mohammadi, "A Novel Plagiarism Detection Approach Combining BERT-based Word Embedding, Attention-based LSTMs and an Improved Differential Evolution Algorithm," *arXiv preprint arXiv:2305.02374*, 2023.
4. A. Greiner-Petter, M. Fröbe, J. P. Wahle, T. Ruas, B. Gipp, A. Aizawa, and M. Potthast, "Overview of the Plagiarism Detection Task at PAN 2025," *CLEF 2025 Working Notes*, CEUR Workshop Proceedings, vol. 4038, pp. 3575–3585, 2025.

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